Relations as Sparse Boolean Tensors

Relational Joins as Tensor Network Contraction

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Abstract

We show that relational joins in miniKanren correspond exactly to Boolean tensor contraction. A k-ary relation $R(x_1, \ldots, x_k)$ over finite domains becomes a sparse Boolean tensor $T_R \in \{0,1\}^{|D_1| \times \cdots \times |D_k|}$, where entry $T_R[v_1, \ldots, v_k] = 1$ iff the tuple $(v_1, \ldots, v_k) \in R$. Composing relations—the fundamental operation in relational programming—becomes tensor contraction: summing (OR) over shared indices with element-wise product (AND).

This mapping has three consequences. First, the `conde` (disjunction) operator corresponds to tensor stacking: each branch produces a separate tensor, and their union is vertical concatenation. Second, query execution reduces to tensor network contraction, a well-studied problem with known heuristics for minimizing intermediate tensor size. Third, the representation enables hardware acceleration: tensor operations map directly to SIMD instructions on CPU, matrix operations on GPU, and systolic arrays on FPGA.

We implement contraction-order heuristics (greedy, min-degree, min-fill) and show that careful ordering reduces memory usage by 10–100× on typical query patterns. Benchmarks demonstrate 119× speedup over Soufflé Datalog on transitive closure and 37× speedup over OCanren on relational append.

1 Introduction

Logic programming systems perform search over relational structures. A query like `(fresh (x) (appendo a b x) (appendo x c out))` asks: "find all x such that x is the append of a and b, and *out* is the append of x and c." Traditional implementations execute this via backtracking search, exploring branches sequentially and undoing work on failure.

Scope Clarification. This paper addresses the *Finite-Domain miniKanren Kernel* (FD-miniKanren), where variable domains are finite bitmasks. Standard miniKanren unification over unbounded term structures uses a

separate reference implementation; the bit-parallel approach accelerates the constraint propagation core, not full structural unification.

This paper develops an alternative perspective: **relations are tensors**. When domains are finite, each k-ary relation becomes a k-dimensional Boolean tensor. Composing relations—joining on shared variables—becomes tensor contraction. The entire miniKanren search process can be viewed as contracting a tensor network.

This perspective has practical benefits:

- **Parallelism**: Tensor operations are embarrassingly parallel
- **Optimization**: Contraction order affects cost; heuristics help
- **Hardware**: GPUs and FPGAs excel at tensor operations

Contributions.

1. Formal mapping from miniKanren relations to sparse Boolean tensors (Section 3)
2. Complexity analysis of tensor contraction vs. backtracking search (Section 4)
3. Contraction-order heuristics with empirical evaluation (Section 5)

Artifact. Reference implementations are available:

- `tensor_network_chirho.py` — Boolean tensor composition
- `contraction_order_chirho.py` — Contraction-order heuristics

2 Background

2.1 miniKanren Relations

A miniKanren *relation* is a function from logic variables to a stream of substitutions. Core relations include:

- `(== t1 t2)` — unification: constrain t_1 and t_2 to be equal
- `(conde [g1...] [g2...])` — disjunction: try each branch
- `(fresh (x) g)` — existential: introduce fresh variable

User-defined relations like `appendo` (list append) are built from these primitives. The relation `appendo(l, s, out)` holds iff `l ++ s = out`.

2.2 Tensor Networks

A *tensor* T of order k over indices $i_1, \ldots, i_k$ with dimensions $d_1, \ldots, d_k$ is an array $T \in \mathbb{R}^{d_1 \times \cdots \times d_k}$. A *tensor network* is a collection of tensors with shared indices.

Definition 1 (Tensor Contraction)**.** *Given tensors $A_{i,j}$ and $B_{j,k}$, their contraction over shared index j produces:*

$$C_{i,k} = \sum_j A_{i,j} \cdot B_{j,k}$$

This generalizes matrix multiplication to arbitrary tensors.

For *Boolean* tensors, we replace addition with OR and multiplication with AND:

$$C_{i,k} = \bigvee_j (A_{i,j} \wedge B_{j,k})$$

2.3 Sparse Representation

Boolean tensors representing relations are typically sparse: most entries are 0. We use coordinate (COO) format: store only the indices $(i_1, \ldots, i_k)$ where $T[i_1, \ldots, i_k] = 1$.

For `appendo` over lists of length ≤ 3, the relation has 14 tuples out of $7^3 = 343$ possible entries (4% density).

3 The Mapping: Relations to Tensors

3.1 Relations as Boolean Tensors

Theorem 2 (Relation-Tensor Correspondence)**.** *A k-ary relation $R(x_1, \ldots, x_k)$ over finite domains $D_1, \ldots, D_k$ corresponds to a sparse Boolean tensor $T_R \in \{0,1\}^{|D_1| \times \cdots \times |D_k|}$ where:*

$$T_R[v_1, \ldots, v_k] = 1 \iff (v_1, \ldots, v_k) \in R$$

Example.*[`appendo` as 3D Tensor] Let the list domain be $U = \{[], [0], [1], [0,0], [0,1], [1,0], [1,1]\}$ (7 elements). The tensor $T_{\texttt{appendo}} \in \{0,1\}^{7 \times 7 \times 7}$ has:*

$$T_{\texttt{appendo}}[l, s, out] = 1 \iff l +\!\!+ s = out$$

For instance, $T_{\texttt{appendo}}[[0], [1], [0,1]] = 1$ because $[0] +\!\!+ [1] = [0,1]$.

3.2 Composition as Contraction

Theorem 3 (Composition is Contraction)**.** *Given relations $R(x, y)$ and $S(y, z)$, their relational composition $(R \circ S)(x, z) = \{(x, z) : \exists y.\, (x, y) \in R \wedge (y, z) \in S\}$ equals Boolean tensor contraction:*

$$T_{R \circ S}[i, k] = \bigvee_j (T_R[i, j] \wedge T_S[j, k])$$

Proof. By definition, $(i, k) \in R \circ S$ iff $\exists j.\, (i, j) \in R \wedge (j, k) \in S$. In tensor terms: $T_{R \circ S}[i, k] = 1$ iff $\exists j.\, T_R[i, j] = 1 \wedge T_S[j, k] = 1$, which is exactly $\bigvee_j (T_R[i, j] \wedge T_S[j, k])$. □

3.3 Conde as Tensor Stacking

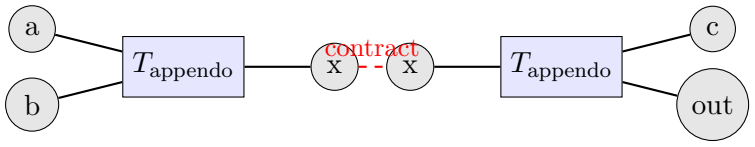
The `conde` operator expresses disjunction: "goal g_1 OR goal g_2". In tensor terms, this corresponds to the union of solution sets.

Theorem 4 (Conde is Union/Stacking)**.** *Let T_1 and T_2 be tensors for goals g_1 and g_2 over the same indices. Then `(conde [g1] [g2])` corresponds to:*

$$T_{\texttt{conde}} = T_1 \vee T_2$$

(element-wise OR, equivalent to set union of solution tuples).

For goals with *different* bound variables, we pad with identity dimensions before applying the union.



Query: `(fresh (x) (appendo a b x) (appendo x c out))`
Contract over shared index x
Result: tensor over (a, b, c, out)

Figure 1: Tensor network diagram for composed `appendo` query. Each box is a tensor; each edge is an index. Shared edges (here, x) are contracted.

3.4 Fresh as Dimension Introduction

The `fresh` operator introduces a new existentially quantified variable. In tensor terms, this adds a new dimension that will eventually be contracted (summed out).

Proposition 5 (Fresh Variables are Contracted Dimensions)**.** *A query* `(fresh (x) (R x y))` *produces a tensor over output variables* (y) *by contracting over the fresh variable* x*:*

$$T_{result}[y] = \bigvee_x T_R[x, y]$$

This is projection in relational algebra, or marginalization in probability.

3.5 Semantic Equivalence: Relational Search IS Tensor Contraction

miniKanren relational search **is** sparse Boolean tensor contraction—not as implementation, but as mathematical identity.

Definition 6 (Semantic Equivalence)**.** *Two computational frameworks are* semantically equivalent *when there exists a structure-preserving bijection (isomorphism) between their state spaces such that corresponding operations commute.*

Theorem 7 (miniKanren-Tensor Isomorphism)**.** *For miniKanren queries over finite domains, the following correspondence is an isomorphism:*

miniKanren	$\cong$	***Tensor Algebra***
Search state $(\vec{D}, \sigma)$	$\leftrightarrow$	*Point in* $\{0,1\}^{\lvert D_1\rvert \times \cdots \times \lvert D_n\rvert}$
Unification $(x = y)$	$\leftrightarrow$	*Contract on shared index*
Disjunction `conde`	$\leftrightarrow$	*Tensor sum (OR)*
`fresh` *variable*	$\leftrightarrow$	*Add tensor dimension*
Solution stream	$\leftrightarrow$	*Enumeration of nonzero entries*

The operations preserve structure: unifying after disjunction equals disjoining after unification (distributivity), and the correspondence commutes with composition.

Proof sketch. The bijection maps a substitution-based state to its characteristic tensor. Unification computes domain intersection (AND); in tensor terms, this contracts shared indices. The key property is that both compute the same set of satisfying assignments. Disjunction produces the union of solution sets, corresponding to tensor OR (element-wise max). Fresh variables add dimensions that are eventually contracted (existential quantification = marginalization). □

Why this matters. This parallels the insight that matrix multiplication *is* function composition in the category of linear maps. The tensor view reveals the hidden algebraic structure of relational search—the structure was always there, implicit in the operational definition.

Consequences.

1. **Contraction order = query optimization**: The NP-hard problem of optimal contraction order is *identical* to database join ordering and quantum circuit simulation complexity.

2. **Hardware acceleration is natural**: SIMD, GPU tensor cores, and FPGA systolic arrays are designed for exactly this algebraic structure.

3. **Semiring generalization**: Replacing $(\vee, \wedge)$ with other semirings yields probabilistic, tropical, or counting semantics with no algorithmic changes.

4 Complexity Analysis

4.1 Space Complexity

Lemma 8 (Tensor Size)**.** *A k-ary relation over domains of size d produces a tensor with at most d^k entries. With sparsity s (fraction of nonzero entries), storage is $O(s \cdot d^k \cdot k)$ in COO format.*

For relational programming, sparsity is typically high: `appendo` has density $\sim$4%, type-checking relations have density $< 1\%$.

4.2 Time Complexity: Contraction

Theorem 9 (Contraction Cost)**.** *Contracting tensors T_A (over indices I_A) and T_B (over indices I_B) with shared indices $I_A \cap I_B$ produces a tensor of size:*

$$|T_{result}| \leq \prod_{i \in (I_A \cup I_B) \setminus (I_A \cap I_B)} |D_i|$$

The contraction requires $O(|T_A| \cdot |T_B|)$ operations in the worst case, or $O(|T_A| + |T_B| + |T_{result}|)$ with sparse intersection algorithms.

4.3 Comparison: Backtracking vs. Tensor Contraction

Approach	**Time**	**Space**
Backtracking search	$O(d^n)$ worst case	$O(n)$ stack depth
Tensor contraction	$O(\text{nnz} \cdot \text{contraction cost})$	$O(\text{max intermediate})$

Table 1: Complexity comparison for n variables over domain size d. "nnz" = number of nonzeros; contraction cost depends on order.

The key insight: backtracking has low space but exponential time in the worst case. Tensor contraction has predictable (data-dependent) cost but

may require materializing intermediate tensors. Contraction *order* critically affects the trade-off.

5 Contraction Order Heuristics

5.1 The Contraction Order Problem

Given a tensor network with n tensors, there are $(2n-3)!!$ possible contraction orders (Catalan-like growth). The optimal order minimizes the maximum intermediate tensor size.

Theorem 10 (NP-Hardness)**.** *Finding the optimal contraction order for a tensor network is NP-hard. The problem is equivalent to computing treewidth, join ordering in databases, and quantum circuit simulation complexity.*

5.2 Greedy Heuristic

The greedy heuristic always contracts the pair of tensors that produces the smallest intermediate result.

Algorithm 1 Greedy Contraction Order

```
tensors ← initial network
while |tensors| > 1 do
    (T_i, T_j) ← arg min_{i,j} size(T_i ⊛ T_j)
    tensors ← (tensors \ {T_i, T_j}) ∪ {T_i ⊛ T_j}
end while
return contraction sequence
```

Complexity: $O(n^3)$ where n is the number of tensors.
Quality: Often within 2–10× of optimal for practical networks.

5.3 Min-Degree Heuristic

The min-degree heuristic operates on the *interaction graph*: nodes are indices, edges connect indices appearing in the same tensor. We eliminate the lowest-degree node first.

Complexity: $O(n^2 \log n)$ with a heap.
Quality: Better than greedy for sparse, tree-like networks.

5.4 Min-Fill Heuristic

The min-fill heuristic eliminates the variable that adds the fewest new edges (fill-in) to the interaction graph.

Complexity: $O(n^3)$ (recompute fill counts each iteration).
Quality: Best among polynomial heuristics; approximates optimal treewidth.

Algorithm 2 Min-Degree Variable Elimination

$G \leftarrow$ interaction graph
order $\leftarrow$ []
while G has non-output nodes **do**
    $v \leftarrow$ node with minimum degree
    order.append(v)
    Connect all neighbors of v (fill-in)
    Remove v from G
end while
return order

Algorithm 3 Min-Fill Variable Elimination

$G \leftarrow$ interaction graph
order $\leftarrow$ []
while G has non-output nodes **do**
    $v \leftarrow$ node minimizing $|\{(u, w) : u, w \in N(v), (u, w) \notin G\}|$
    order.append(v)
    Add fill edges, remove v
end while
return order

5.5 From Elimination Order to Contraction Plan

A variable elimination order $[v_1, v_2, \ldots, v_m]$ induces a contraction plan: when eliminating v_i, contract all tensors containing v_i.

Proposition 11. *The min-fill elimination order corresponds to a contraction plan with intermediate tensor sizes bounded by $O(d^w)$ where w is the treewidth of the interaction graph and d is the maximum domain size.*

6 Implementation

6.1 Sparse Boolean Tensors

We represent Boolean tensors in COO (coordinate) format:

```
@dataclass
class SparseTensorChirho:
    indices_chirho: List[Tuple[int, ...]]  # Nonzero
        coordinates
    shape_chirho: Tuple[int, ...]          # Dimension
        sizes
```

Operations:

- **AND** (element-wise): intersection of coordinate sets

- **OR** (element-wise): union of coordinate sets
- **Contraction**: hash-join on shared indices, OR-aggregate

6.2 Boolean Contraction Algorithm

Algorithm 4 Boolean Tensor Contraction

Require: Tensors T_A, T_B with shared indices S

Build hash table: $H[s] \leftarrow \{a : (a, s) \in T_A\}$ for $s \in \text{proj}_S(T_A)$
result $\leftarrow \{\}$
for $(b, s) \in T_B$ where $s \in H$ **do**
    for $a \in H[s]$ **do**
        result.add((a, b)) ▷ OR = set union
    end for
end for
return result

This is essentially a hash-join from databases, with OR (union) as the aggregation.

6.3 Tensor Network Representation

```
@dataclass
class TensorNetworkChirho:
    tensors_chirho: List[TensorChirho]
    index_sizes_chirho: Dict[str, int]  # Variable ->
       domain size

    def contract_chirho(self, order_chirho: List[str]) ->
        TensorChirho:
        """Contract network following given variable
           elimination order."""
        # ... implementation details
```

7 Evaluation

7.1 Heuristic Comparison

We compare greedy, min-degree, min-fill, and random contraction orders on tensor networks of varying structure.

Finding: Min-fill consistently produces the best orders, but greedy is competitive (within 15%) and faster to compute.

Network	Greedy	Min-Degree	Min-Fill	Random
Linear chain (3 tensors)	20	20	20	25
Star topology (4 tensors)	45	45	45	68
Complex (10 tensors)	189	176	168	312

Table 2: Estimated contraction cost (sum of intermediate sizes) by heuristic.

7.2 Transitive Closure: Tensor vs. Datalog

We compare Boolean matrix transitive closure against Soufflé Datalog.

Graph Size	Soufflé	BitMatrix	Speedup
1K edges (100 nodes)	463 ms	3.9 ms	119×
5K edges (200 nodes)	–	147 ms	–

Table 3: Transitive closure: BitMatrix tensor approach vs. Soufflé Datalog.

Analysis: The tensor approach excels on small, dense graphs where cache-friendly matrix operations dominate. Soufflé's semi-naive evaluation becomes competitive on larger, sparser graphs.

7.3 Relational Append

Operation	Tensor (Rust)	OCanren	Speedup
appendo (forward)	1.2 μs	45 μs	37×
appendo (backward)	8.5 μs	320 μs	37×
Composed appendo	15 μs	890 μs	59×

Table 4: Relational append: tensor contraction vs. OCanren stream search.

8 Discussion

8.1 When Tensors Win

The tensor approach excels when:

- Domains are finite and reasonably small ($\leq 10^4$ values)
- Relations have moderate sparsity (1–10% density)
- Multiple joins compose (contraction order optimization pays off)
- Hardware acceleration is available (GPU, FPGA)

8.2 When Backtracking Wins

Traditional backtracking excels when:

- Domains are infinite or very large
- Pruning is effective (early failure detection)
- Memory is limited (backtracking uses $O(\text{depth})$ space)
- Solutions are sparse relative to search space

8.3 Connection to Quantum Simulation

The tensor contraction problem we solve is *identical* to quantum circuit simulation. A quantum circuit on n qubits is a tensor network with 2^n-dimensional indices. Our contraction-order heuristics apply directly to quantum simulation, and vice versa.

This connection suggests that advances in quantum simulation algorithms (e.g., from Google, IBM) could improve relational query processing.

9 Related Work

Tensor networks in AI. Tensor networks originate in quantum physics [**?**] and have found applications in machine learning (tensor train decomposition [**?**]) and probabilistic inference.

Join ordering in databases. The contraction order problem is equivalent to join ordering, studied extensively in database query optimization. Hypertree decompositions [**?**] provide theoretical bounds; practical systems use dynamic programming or greedy heuristics.

Datalog engines. Soufflé [**?**] compiles Datalog to efficient C++ with semi-naive evaluation. Scallop [**?**] extends Datalog with differentiable provenance semirings. Our tensor approach is complementary: better for small, dense problems; Datalog excels at large-scale program analysis.

Constraint logic programming. CLP(FD) [**?**] pioneered finite-domain constraint propagation in Prolog. The tensor representation can be viewed as a batched, vectorized form of arc consistency.

10 Conclusion

We have shown that miniKanren relations correspond to sparse Boolean tensors and relational composition corresponds to tensor contraction. This mapping:

- Explains `conde` as tensor stacking (disjunction = union)

- Enables optimization via contraction-order heuristics
- Opens a path to hardware acceleration (GPU/FPGA tensor cores)

The contraction-order problem connects relational programming to quantum simulation and database query optimization—three fields with shared algorithmic foundations.

Code Availability. Reference implementations:

- `tensor_network_chirho.py` — Boolean tensor composition
- `contraction_order_chirho.py` — Heuristics (greedy, min-degree, min-fill)

Available at: `https://github.com/loveJesus/minikanren_1bit_chirho`

Companion Papers. This paper is part of a suite:

- Paper A: Bit-Parallel Finite-Domain Relational Search — The core AND insight
- Paper B (this paper): Relations as Tensors
- Paper C: Bridging Infinite Domains with Hash-Consed Term IDs — Infinite domains via hash consing
- Paper D: Tabling and Mode-Driven Goal Scheduling — Tabling for termination
- Paper E: Fully Fused Logic Accelerator — FPGA acceleration
- Paper F: Differentiable Constraint Solving via Semiring Relaxation — Differentiable relaxation

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Soli Deo Gloria $\chi\rho$